

- The Intersection of P and Q is the multiset in which the multiplicity of an element is the minimum of its multiplicities in P and Q . ($P \cap Q$)
- The Difference of P and Q ($P - Q$) is the multiset in which the multiplicity of an element is the multiplicity of the element in P less than its multiplicity in Q unless this difference is negative, in which the multiplicity is 0.
- The Sum of P and Q is the multiset in which the multiplicity of an element is the sum of multiplicities in P and Q . ($P + Q$).

Example 21: Suppose that P and Q are the multisets $\{4 \cdot a, 1 \cdot b, 3 \cdot c\}$ and $\{3 \cdot a, 4 \cdot b, 2 \cdot d\}$, respectively. Find $P \cup Q$, $P \cap Q$, $P - Q$ and $P + Q$.

Solution: $P \cup Q = \{\max(4, 3) \cdot a, \max(1, 4) \cdot b, \max(3, 0) \cdot c, \max(0, 2) \cdot d\}$
 $= \{4 \cdot a, 4 \cdot b, 3 \cdot c, 2 \cdot d\}$

$$P \cap Q = \{\min(4, 3) \cdot a, \min(1, 4) \cdot b, \min(3, 0) \cdot c, \min(0, 2) \cdot d\}$$

$$= \{3 \cdot a, 1 \cdot b, 0 \cdot c, 0 \cdot d\} = \{3 \cdot a, 1 \cdot b\}$$

$$P - Q = \{(4 - 3) \cdot a, (1 - 4) \cdot b, (3 - 0) \cdot c, (0 - 2) \cdot d\}$$

$$= \{1 \cdot a, 0 \cdot b, 3 \cdot c, 0 \cdot d\} = \{1 \cdot a, 3 \cdot c\}$$

$$P + Q = \{(4 + 3) \cdot a, (1 + 4) \cdot b, (3 + 0) \cdot c, (0 + 2) \cdot d\} = \{7 \cdot a, 5 \cdot b, 3 \cdot c, 2 \cdot d\} \square$$